

Further applications of the normal distribution



- 1
a 44% b 89% c 50 848
- 2
a 23% b 13% c 2524
- 3
a 0.769 b 70.060% c 31.757%
- 4
a 26.154 cm b 27.280 cm c 24.724 cm d 0.248
e 0.690
- 5
a 3.493 kg b 3.721 kg c 3.363 kg d 0.309
e 0.614
- 6
a 190.047 cm b 195.869 cm c 4.8% d 0.566
e 0.371
- 7
a 5.108, 5.012
b 0.3
c There is a good chance that the machine is faulty.
- 8
a 301.15 K b 3.3 K
- 9
a 3500 g b 550 g
- 10
a 550 mm b 1600 mm
- 11
a $b = 4$ b 68.48

12

a -328



b 16

13

a 2.985

b 67.848 g

14

a 34.457 cm

b $(X + 11)$ cm

c 34.457 cm

15

$$a \quad 0.706 = \frac{70 - \mu}{\sigma}$$

$$b \quad -0.151 = \frac{56 - \mu}{\sigma}$$

$$c \quad \mu = 58.465, \sigma = 16.331$$

16

a 58.7%

b $(0.7\mu + 2)\%$ c $(0.7\sigma)\%$

17

a 122.736

$$b \quad \frac{20}{37}X$$

18

a 40%

b 3

c 69%

19

a 40.1

b 0.69

c 0.035

d 0.346

e 1.74 mins

20

a 83%

b 0.64

c 44%

21

a 0.43

b 172.83 g

c 0.85

d 177.26 g